

Semester Test 2

FSK176

7 October/Oktobor 2009

Semestertoets 2

Surname and Initials: _____

Van en voorletters:

Student Number: _____

Studentenommer:

Signature: _____

Handtekening:

Time : 2½ hours

Total : 70

Tyd:

Totaal:

Answer all questions. Clearly show all calculations. Do not just write down the answer. Where applicable draw the free-body diagrams. Programmable calculators may NOT be used. No textbooks allowed. A formula sheet is attached at the end of this paper. All test and examination regulations of the University of Pretoria are applicable	<i>Beantwoord alle vrae.</i> <i>Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.</i> <i>Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).</i> <i>Programmeerbare sakrekenaars MAG NIE gebruik word nie.</i> <i>Geen handboeke word toegelaat nie.</i> <i>'n Formuleblad is agter aan die vraestel aangeheg.</i> <i>Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.</i>
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SEMESTER TEST 2

FSK 176

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7 October/Oktobor 2009

Surname and Initials: _____

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Lokaal:

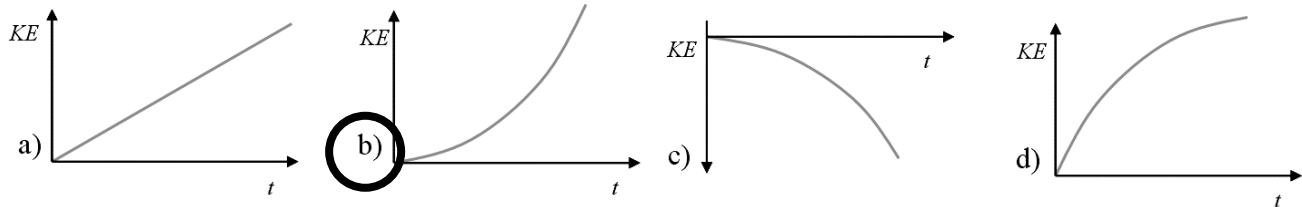
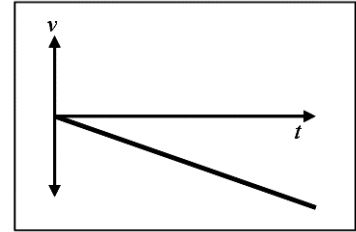
Question 1 / Vraag 1**[3]**

Clearly circle your choice for the correct answer. Please use a pen to indicate your choice.

Trek 'n kring om jou keuse vir die korrekte antwoord. Gebruik asseblief 'n pen om jou keuse aan te dui.

- 1.1 The graph alongside shows the velocity of an object as a function of time. Which of the graph of kinetic energy versus time matches the velocity data?

Die grafiek langsaaan dui die snelheid van 'n voorwerp as 'n funksie van tyd. Watter een van die kinetiese energie teen tyd grafieke onderaan stem ooreen met die snelheids data.



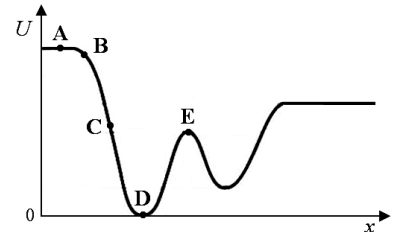
- 1.2 Complete the following statement: A force that acts on an object is said to be conservative if

Voltooi die volgende stelling: 'n Krag wat toegepas word op 'n voorwerp is konserwatief indien

- a) the work it does on the object is equal to the increase in the object's kinetic energy
die arbeid wat dit verrig op die voorwerp gelyk is aan die toename in die voorwerp se kinetiese energie
- ☒ b) the work it does on the object is independent of the path of the motion.
die arbeid wat dit verrig op die voorwerp onafhanklik is van die bewegingspad.
- c) it always acts in the direction of motion of the object.
dit altyd toegepas word in die rigting van die beweging van die voorwerp.
- d) it always results in a change in the object's kinetic energy.
dit altyd 'n verandering in die voorwerp se kinetiese energie veroorsaak.
- e) it obeys Newton's laws of motion.
dit Newton se wette van beweging gehoorsaam.

- 1.3 The graph shows the potential energy as a function of distance for an object moving along the x axis. At which of the labeled points does the force acting on the object have the largest magnitude?

Die grafiek dui die potensiële energie as 'n funksie van die afstand wat die voorwerp beweeg langs die x as. By watter van die punte aangedui is die grootte van die krag op die voorwerp die hoogste waarde?



- a) A
- b) B
- ☒ c) C
- d) D
- e) E

- 1.4 Momentum may be expressed using which of the following units?

Watter van die volgende kan gebruik word vir die eenheid van momentum?

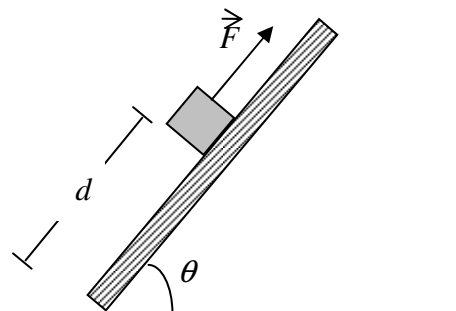
- a) N / s
- b) kg / m
- ☒ c) kg · m/s
- d) kg / (m · s)
- e) N · kg / s

- 1.5 Two objects are involved in an elastic collision. Which one of the following statements concerning this situation is false?
Twee voorwerpe is betrokke in 'n elastiese botsing. Watter een van die volgende stellings in verband met die situasie is vals.
- a) The total momentum is conserved.
Die totale momentum bly behoue.
 - b) The magnitude of the force exerted by each object on the other object is equal.
Die grootte van die eksterne krag wat elke voorwerp op die ander oefen is gelyk.
 - ☒ c) The kinetic energy of each object is the same before and after the collision.
Die kinetiese energie van elke voorwerp is dieselfde voor en na die botsing.
 - d) The total kinetic energy before the collision is equal to the total kinetic energy after the collision.
Die totale kinetiese energie voor die botsing is gelyk aan die totale kinetiese energie na die botsing.
 - e) The total kinetic energy is conserved.
Die totale kinetiese energie bly behoue.
- 1.6 While putting in a new ceiling, Jake uses a drill to put screws in. The screws rotate clockwise as they go into the ceiling. What is the direction of the angular velocity of the screw as the drill drives it into the ceiling? Express the direction relative to Jake, who is looking upward at the screw.
Jake gebruik 'n boormasjien om skroewe in die plafon in te draai. Die skroewe roteer kloksgewys soos hulle ingaan. Wat is die rigting van die hoeksnelheid van die skroewe soos die boor hulle indraai? Gee jou antwoord ten opsigte van Jake, wie van onder opkyk na die skroewe.
- a) down
af
 - ☒ b) up
op
 - c) left
links
 - d) right
regs
 - e) parallel to the ceiling
parallel met die plafon
- 1.7 Which one of the following statements concerning the rotational inertia is false?
Watter van die volgende stellings met betrekking tot die traagheidsmoment is nie waar nie?
- ☒ a) The rotational inertia depends on the angular acceleration of the object as it rotates.
Die traagheidsmoment is afhanklik van die hoekversnelling van die voorwerp wat roteer.
 - b) The rotational inertia may be expressed in units of $\text{kg} \cdot \text{m}^2$.
Die traagheidsmoment se eenheid kan $\text{kg} \cdot \text{m}^2$ wees.
 - c) The rotational inertia depends on the orientation of the rotation axis relative to the particles that make up the object.
Die traagheidsmoment hang af van die orientasie van die rotasie as relatief tot die deeltjies waaruit die voorwerp bestaan.
 - d) Of the particles that make up an object, the particle with the smallest mass may contribute the greatest amount to the rotational inertia.
Indien die deeltjies waaruit die voorwerp gemaak is, beskou word, is dit moontlik dat die deeltjie met die kleinste massa, die meeste kan bydra tot die traagheidsmoment.
 - e) The rotational inertia depends on the location of the rotation axis relative to the particles that make up the object.
Die traagheidsmoment hang af van die rotasie-as se posisie, relatief tot die deeltjies waaruit die voorwerp bestaan.

Question 2 /Vraag 2

[2+2+2+2 = 8]

A block of ice slides down a frictionless ramp at angle $\theta = 50^\circ$ while a worker pulls on the block (via a rope) with a force $\vec{F}$ that has a magnitude of 50N and is directed up the ramp. As the block slides through a distance $d = 0.50$ m along the ramp, its kinetic energy increases by 80 J.

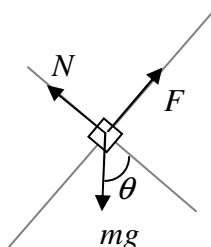


- Determine the net force on the ice block.
- Determine the mass of the block.
- How much greater would its kinetic energy have been if the rope had no been attached?

'n Blok ys gly langs 'n wrywinglose sluins vlak af terwyl 'n werker die

blok met 'n tou trek met 'n krag $\vec{F}$, met grootte van 50N gerig opwaarts langs die skuinste. Die hoek $\theta = 50^\circ$. Soos die blok oor 'n afstand $d = 0.50$ m by die skuinste afgly, styg sy kinetiese energie met 80 J.

- Bepaal die netto krag op die ysblok.
- Bepaal die massa van die blok.
- Hoeveel meer sou die kinetiese energie van die blok gewees het indien dit nie aan die tou vasgemaak was nie?



✓✓

a)

$$\Delta K = W = F \cdot d = Fd \cos \theta = Fd$$

$$F = \frac{\Delta K}{d} = \frac{80 \text{ J}}{0.5 \text{ m}} = 160 \text{ N}$$

✓✓

down the slope

$$F_{\text{net}} = F - mg \sin \theta$$

$$\text{b) } m = -\frac{F_{\text{net}} - F}{g \sin \theta} = \frac{-160 - 50}{-9.8 \sin 50} = 27.9$$

✓✓

$$= 28 \text{ kg}$$

c) Work done by applied force:

$$W = F \cdot d = Fd \cos 180^\circ$$

$$= -50 \text{ N}(0.5 \text{ m})$$

✓

$$= -25 \text{ J}$$

Or net change in KE

$$\Delta K = F_{\text{net}} d \cos \theta$$

$$= (mg \sin 50^\circ) d = 28 \text{ kg}(9.8 \text{ m/s}^2) \sin 50^\circ (0.5 \text{ m})$$

✓

$$= 105 \text{ J}$$

Kinetic energy would have been 25 J more. ✓

Question 3 / Vraag 3**[2+1+1 = 4]**

A 1.50 kg snowball is fired from a cliff 12.5 m high. The snowball's initial velocity is 14.0 m/s, directed 41.0° above the horizontal.

- How much work is done on the snowball by the gravitational force during its flight to the flat ground below the cliff?
- What is the change in the gravitational potential energy of the snowball – Earth system during the flight?
- If the gravitational potential energy is taken to be zero at the height of the cliff, what is its value when the snowball reaches the ground?

'n 1.5 kg sneeubal word van 'n 12.5 m hoë krans gevuur. Die sneeubal se beginsnelheid is 14.0 m/s gerig 41.0° bo die horisontaal.

- Hoeveel arbeid word verrig op die sneeubal deur die gravitasiekrag gedurende die vlug vanaf die krans na die gelyk grond?
- Wat is die verandering in die gravitasie potensiële energie van die sneeubal-Aarde sisteem tydens die vlug?
- Indien die gravitasie potensiële energie nul is bo op die krans, wat is die waarde daarvan wanneer die sneeubal die grond bereik?

a)

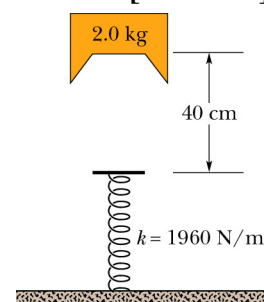
$$\begin{aligned}
 W &= \vec{F} \cdot d = mgd \cos \phi \\
 &= (1.5 \text{ kg})(9.8 \text{ m/s}^2)(12.5 \text{ m}) \cos 0^\circ \quad \checkmark \checkmark \\
 &= 183.75 \\
 &\approx 184 \text{ J}
 \end{aligned}$$

b) $\Delta U = -W = -184 \text{ J} \quad \checkmark$ c) $U = -184 \text{ J} \quad \checkmark$ **Question 4 / Vraag 4****[3+3+2=8]**

A block of mass $m = 2.0 \text{ kg}$ is dropped from a height $h = 40 \text{ cm}$ onto a spring of spring constant $k = 1960 \text{ N/m}$ (See figure).

- Find the velocity of the block when it hits the spring.
 - Find the maximum distance the spring is compressed.
 - Determine the work done on the block by the spring.
- 'n Blok met 'n massa $m = 2.0 \text{ kg}$ word van 'n hoogte $h = 40 \text{ cm}$ gelos en val op 'n veer met veerkonstante $k = 1960 \text{ N/m}$ (Sien figuur).

- Bepaal die snelheid van die blok wanneer dit die veer tref.
- Bepaal die maksimum afstand waardeur die veer saamgepers word.
- Bepaal die arbeid verrig deur die veer op die blok.

a) Let the reference point be the top of the spring. i.e. $U = 0 \text{ J}$ at top of spring. $\checkmark$

$$(K + U)_i = (K + U)_f$$

$$(0 + mgh) = \left(\frac{1}{2}mv^2 + 0\right)$$

$$\begin{aligned}
 v &= \sqrt{2mgh / m} = \sqrt{2gh} \quad \checkmark \checkmark \\
 &= \sqrt{2(9.8 \text{ m/s}^2)(0.4 \text{ m})} \\
 &= 2.8 \text{ m/s}
 \end{aligned}$$

b) KE at top of spring = PE of spring at full compression. $\checkmark$

$$\Delta K = \Delta U_{spring}$$

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

$$x = \sqrt{\frac{mv^2}{k}} = \sqrt{\frac{(2.0 \text{ kg})(2.8 \text{ m/s})^2}{1960 \text{ N/m}}} \checkmark \checkmark$$

$$= 0.089 \text{ m}$$

$$\approx 0.09 \text{ m}$$

c)

$$W = \int_0^x F dx = \int_0^x \frac{1}{2}kx dx = \frac{1}{2}kx^2$$

$$= \frac{1}{2}(1960 \text{ N/m})(0.09 \text{ m})^2 \checkmark \checkmark$$

$$= 7.94 \text{ J}$$

Question 5 /Vraag 5

[3]

How far is the centre of mass of the Earth Moon system from the centre of the earth? Give your final answer as a multiple or fraction of the Earth's radius.

Hoe ver is die swaartepunt van die Aarde-Maan sisteem van die middelpunt van die Aarde? Druk jou finale antwoord in terme van die Aarde se radius uit.

Let earth be at the origin of the system: $x_E = 0 \text{ m}$

Let moon be a distance: x_M from Earth, (the distances given in formula sheet is centre to centre distance)

$$x = \frac{1}{m_E + m_M} [m_E x_E + m_M x_M]$$

$$= \frac{1}{5.98 \times 10^{24} \text{ kg} + 7.36 \times 10^{22} \text{ kg}} [(5.98 \times 10^{24} \text{ kg})0 + (7.36 \times 10^{22} \text{ kg})(3.82 \times 10^8 \text{ m})] \checkmark \checkmark$$

$$= 4644377 \text{ m}$$

$$= 4.64 \times 10^6 \text{ m}$$

Radius of Earth: $6.37 \times 10^6 \text{ m}$

$$x = 0.73r_E \checkmark$$

Question 6 / Vraag 6**[3+3=6]**

A certain radioactive (parent) nucleus transforms to a different (daughter) nucleus by emitting an electron and a neutrino. The parent nucleus was at rest at the origin of an xy coordinate system. The electron moves away from the origin with linear momentum $(-1.2 \times 10^{-22} \text{ kg}\cdot\text{m/s})\hat{i}$; the neutrino moves away from the origin with linear momentum $(-6.4 \times 10^{-23} \text{ kg}\cdot\text{m/s})\hat{j}$.

a) Determine the magnitude and direction of the linear momentum of the daughter nucleus?

b) If the daughter nucleus has a mass of $5.8 \times 10^{-26} \text{ kg}$, what is its kinetic energy?

'n Radioaktiewe kern verval na 'n ander (dogter) kern deur 'n elektron en 'n neutrino af te gee. Die oorspronklike kern was aanvanklik in rus by die oorsprong van 'n xy koördinaatstelsel. Die elektron beweeg weg van die oorsprong met 'n lineêre momentum van $(-1.2 \times 10^{-22} \text{ kg}\cdot\text{m/s})\hat{i}$; die neutrino beweeg weg van die oorsprong met 'n lineêre momentum van $(-6.4 \times 10^{-23} \text{ kg}\cdot\text{m/s})\hat{j}$.

a) Bereken die grootte en rigting van die lineêre momentum van die dogter kern?

Indien die dogter kern 'n massa van $5.8 \times 10^{-26} \text{ kg}$ het, wat is hierdie kern se kinetiese energie?

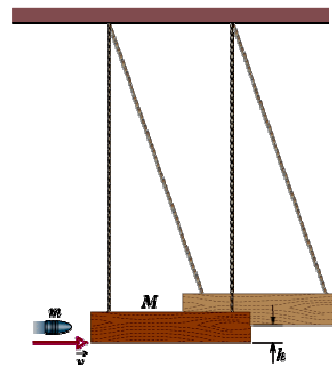
Initial $p = 0$ Initial $p_x = 0$ Initial $p_y = 0$	Final $p = 0 = p_e + p_n + p_d$ Final $p_x = 0 = p_e + p_{dx}$ Final $p_y = 0 = p_n + p_{dy}$
a) $0 = p_e + p_{dx} \qquad 0 = p_n + p_{dy}$ $p_{dx} = -p_e = 1.2 \times 10^{-22} \text{ kg}\cdot\text{m/s} \qquad p_{dy} = -p_n = 6.4 \times 10^{-23} \text{ kg}\cdot\text{m/s}$ $p_d = \sqrt{p_{dx}^2 + p_{dy}^2}$ $= \sqrt{(1.2 \times 10^{-22} \text{ kg}\cdot\text{m/s})^2 + (6.4 \times 10^{-23} \text{ kg}\cdot\text{m/s})^2} \checkmark \checkmark$ $= 1.36 \times 10^{-22} \text{ kg}\cdot\text{m/s}$ Direction: $\theta = \tan^{-1} \left(\frac{6.4 \times 10^{-23}}{1.2 \times 10^{-22}} \right) = 28^\circ$ above the $+x$ axis $\checkmark$	
b) $p = mv \qquad \checkmark$ $v = \frac{p}{m}$ $K = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{p}{m} \right)^2$ $= \frac{p^2}{2m} = \frac{(1.36 \times 10^{-22} \text{ kg}\cdot\text{m/s})^2}{2(5.8 \times 10^{-26} \text{ kg})} = 1.6 \times 10^{-19} \text{ J} \qquad \checkmark \checkmark$	

Question 7 / Vraag 7

[6]

The figure shows a large block of wood of mass $M = 5.5 \text{ kg}$ hanging from two long cords. A bullet of mass $m = 9.0 \text{ g}$ is fired into the block, coming to rest in the block. The block-bullet then swing upward, their centre of mass rising a vertical distance $h = 6.5 \text{ cm}$ before the pendulum comes to rest for a moment. What is the speed of the bullet just prior to the collision?

Die figuur dui 'n groot houtblok aan met massa $M = 5.5 \text{ kg}$ wat aan 2 lang toue hang. 'n Koeël met massa $m = 9.0 \text{ g}$ word na die blok gevuur en kom tot rus in die blok. Die blok-koeël swaai opwaarts en die swaartepunt styg met 'n vertikale afstand $h = 6.5 \text{ cm}$ voordat die pendulum tot stilstand kom. Wat is die spoed van die koeël net voor die botsing?



$$p_i = p_f$$

$$mv_i = (M + m)v_f \quad \checkmark \checkmark$$

$$v_i = \frac{M + m}{m} v_f$$

Conservation of E_{mech}

$$\frac{1}{2}(M + m)v_f^2 = (M + m)gh \quad \checkmark \checkmark$$

$$v_f = \sqrt{2gh}$$

$$v_i = \frac{M + m}{m} v_f = \frac{M + m}{m} \sqrt{2gh}$$

$$= \left(\frac{5.5 + 9.0 \times 10^{-3} \text{ kg}}{9.0 \times 10^{-3} \text{ kg}} \right) \sqrt{2(9.8 \text{ m/s}^2)(6.5 \times 10^{-2} \text{ m})} \quad \checkmark \checkmark$$

$$= 691 \text{ m/s}$$

Question 8 / Vraag 8

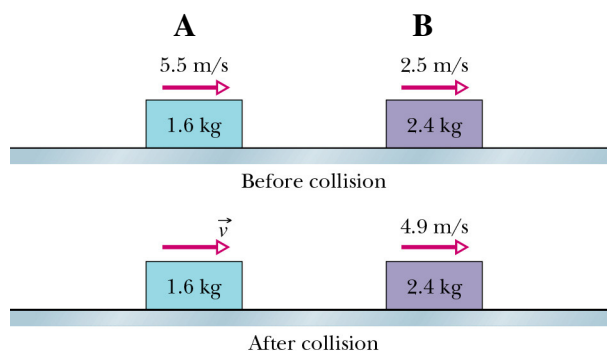
[3+3+3=9]

In the figure, block A (mass 1.6 kg) slides into block B (mass 2.4 kg) along a frictionless surface. Three velocities are indicated in the figure.

- What is the velocity of block A after the collision?
- Indicate if the collision is elastic.
- Suppose the velocity of block B is initially in the opposite direction to that shown. Can the velocity of block A after the collision be in the direction shown in the figure?

In die figuur, blok A (massa 1.6 kg) gly oor 'n wrywinglose oppervlak en bots met blok B (massa 2.4 kg). Drie snelhede word aangedui in die figuur.

- Wat is die snelheid van blok A na die botsing?
- Toon aan of die botsing elasties is.
- Veronderstel dat blok B in die teenoorgestelde rigting beweeg voor die botsing. Kan die snelheid van blok A na die botsing in dieselfde rigting wees soos aangedui in die figuur?



a)

$$p_i = p_f$$

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

$$v_{Af} = \frac{m_A v_{Ai} + m_B v_{Bi} - m_B v_{Bf}}{m_A} \quad \checkmark \checkmark \checkmark$$

$$= \frac{1.6(5.5) + 2.4(2.5) - 2.4(4.9)}{1.6}$$

$$= 1.9 \text{ m/s}$$

b)

$$K_i = \frac{1}{2}m_A v_{Ai}^2 + \frac{1}{2}m_B v_{Bi}^2$$

$$= \frac{1}{2}(1.6)(5.5)^2 + \frac{1}{2}(2.4)(2.5)^2 = 31.7 \text{ J} \quad \checkmark \checkmark$$

$$K_f = \frac{1}{2}m_A v_{Af}^2 + \frac{1}{2}m_B v_{Bf}^2$$

$$= \frac{1}{2}(1.6)(1.9)^2 + \frac{1}{2}(2.4)(4.9)^2 = 31.7 \text{ J}$$

KE before = KE after therefore Collision is elastic $\checkmark$

c)

$$p_i = p_f$$

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

$$v_{Af} = \frac{m_A v_{Ai} - m_B v_{Bi} - m_B v_{Bf}}{m_A} \quad \checkmark \checkmark$$

$$= \frac{1.6(5.5) - 2.4(2.5) - 2.4(4.9)}{1.6}$$

$$= -5.6 \text{ m/s}$$

Direction of velocity after collision will be in the opposite direction to that indicated. $\checkmark$

Question 9 /Vraag 9**[2+3+3+3 = 11]**

A magazine article describes a trolley bus in Zürich which uses a flywheel as energy source. At each stop, the angular velocity of the wheel is increased so that it can reach the following stop. The flywheel is in the form of a solid cylinder with mass 1000 kg and diameter 1.8 m rotating about the central axis.

- Calculate the moment of inertia of the wheel.
- The maximum angular velocity that the fly-wheel reaches is 3000 revolutions per minute. Calculate the radial acceleration of a point on the outside of the wheel at this angular velocity.
- Calculate the kinetic energy of the wheel at this angular velocity.
- The average power needed to drive the trolley bus is 1.85×10^4 W. Calculate how long the bus can ride if the fly-wheel has maximum angular velocity. Give your final answer in minutes.

'n Tydskrifartikel beskryf 'n trolliebus in Zürich wat 'n vliegwiel gebruik as sy energiebron. By elke halte word die hoeksnelheid van die wiel verhoog sodat dit die volgende halte kan bereik. Die vliegwiel is in die vorm van 'n soliede silinder met massa 1000 kg en deursnee 1.8 m wat om sy sentrale as roteer.

- Bereken die traagheidsmoment van die vliegwiel.
- Die maksimum hoeksnelheid wat die vliegwiel bereik is 3000 omwentelinge per minuut. Bereken die radiale versnelling van 'n punt op die buitekant van die wiel by hierdie hoeksnelheid.
- Bereken die kinetiese energie wat die wiel besit by hierdie hoeksnelheid.
- Die gemiddelde drywing benodig om die trolliebus te dryf is 1.85×10^4 W. Bereken hoe lank dit kan ry as die vliegwiel sy maksimum hoeksnelheid het. Gee jou finale antwoord in minute.

a)	$I = \frac{1}{2}MR^2 = \frac{1}{2}(1000)(0.9)^2$ ✓✓ $= 405 \text{ kg m}^2$
b)	$a_r = \omega^2 r = \left(\frac{3000 \times 2\pi}{60}\right)^2 (0.9)$ ✓✓✓ $= 88826 \text{ m / s}^2$
c)	$K = \frac{1}{2}I\omega^2$ $= 0.5(405)\left(\frac{3000 \times 2\pi}{60}\right)^2 = 2.0 \times 10^7 \text{ J}$ ✓✓✓
d)	$W = \Delta K$ $P = \frac{W}{t}$ $t = \frac{K}{P} = 2 \times 10^7 / (1.85 \times 10^4) = 1080 \text{ s}$ ✓✓✓ $= 18 \text{ min}$

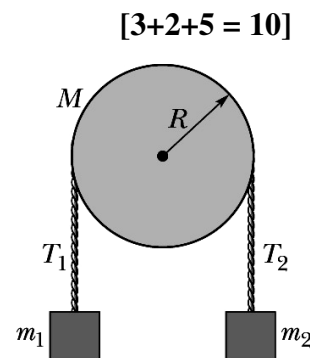
Question 10

In the figure, two blocks, of mass $m_1 = 400 \text{ g}$ and $m_2 = 600 \text{ g}$ are connected by a massless cord that is wrapped around a uniform disk of mass $M = 500 \text{ g}$ and radius $R = 12.0 \text{ cm}$ (It is not a pulley). The disk can rotate without friction about a fixed horizontal axis through its centre; the cord cannot slip on the disk. The system is released from rest.

- Determine the moment of inertia of the disk.
- Draw the freebody diagrams and using both forms of Newton's second law, determine the magnitude of the acceleration of the blocks.

In die figuur is twee blokke, met massas $m_1 = 400 \text{ g}$ en $m_2 = 600 \text{ g}$, gekonnekteer deur 'n tou wat om 'n uniforme skyf van massa $M = 500 \text{ g}$ en radius $R = 12.0 \text{ cm}$ gedraai is (Dis nie 'n "pulley"/katrol nie). Die skyf kan sonder wrywing om 'n horisontale as deur die middelpunt roteer; die tou kan nie op die skyf gly nie. Die sisteem word vanaf 'n toestand van rus losgelaat.

- Bereken die traagheidsmoment van die skyf.
- Teken die vryliggaamsdiagramme en deur gebruik te maak van albei vorms van Newton se wet, bepaal die grootte van die versnelling van die blokke.



Freebody diagrams / Kragtediagramme:

✓	✓	✓

a)

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(0.5 \text{ kg})(0.12 \text{ m})^2 = 0.0036 \text{ kg} \cdot \text{m}^2$$

✓✓

b) In this question, we cannot say $T_1 = T_2$ (as we did previously in ch 5 and 6) since the disc has a significant mass and therefore moment of inertia and therefore we cannot ignore its contribution to the system. We have 3 free body diagrams and therefore 3 equations where we apply Newton's 2nd law.

$T_1 - m_1g = m_1a$	- 1
$-T_2 + m_2g = m_2a$	- 2 ✓
$T_1R - T_2R = -I\alpha$	- 3

3/R

$$\frac{T_1R - T_2R}{R} = -\frac{\frac{1}{2}MR^2 \left(\frac{a}{R}\right)}{R}$$

✓

$-T_1 + T_2 = \frac{1}{2}Ma$	- 4
------------------------------	-----

1 + 2 + 4

$$-m_1g + m_2g = m_1a + m_2a + \frac{1}{2}Ma$$

$$(-m_1 + m_2)g = (m_1 + m_2 + \frac{1}{2}M)a$$

$$a = \frac{(-m_1 + m_2)g}{(m_1 + m_2 + \frac{1}{2}M)} = \frac{0.2}{1.25}g \quad \checkmark \checkmark \checkmark$$

$$= 0.16g = 1.57 \text{ m/s}^2$$

Formula sheet:

Mass of Earth: 5.98×10^{24} kg

Mass of Moon: 7.36×10^{22} kg

Mass of Sun: 1.99×10^{30} kg

Radius of Earth: 6.37×10^6 m

Radius of Moon: 1.74×10^6 m

Distance Earth to Moon: 3.82×10^8 m

Distance Earth to Sun: 1.50×10^{11} m

Circle: Circumference $= 2\pi r =$
omtrek

$$\text{Area} = \pi r^2$$

Sphere: Area $= 4\pi r^2$

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Cylinder / Silinder

$$\text{Area} = 2(\pi r^2) + 2\pi rh$$

$$\text{Volume} = \pi r^2 h$$

Trapezium: Area $= \frac{1}{2}(a+b)h$

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\begin{aligned} \vec{a} \times \vec{b} = & (a_y b_z - b_y a_z) \hat{i} \\ & + (a_z b_x - b_z a_x) \hat{j} \\ & + (a_x b_y - b_x a_y) \hat{k} \end{aligned}$$

$$v = v_0 + at$$

$$v_{avg} = \frac{1}{2}(v_0 + v)$$

$$v = v_0 + \frac{1}{2}at$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$x = x_0 + v_{avg} t \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = \frac{v^2}{r}$$

$$g = 9.8 \text{ m/s}^2$$

$$h = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_0 + v)t$$

$$x - x_0 = vt - \frac{1}{2}at^2$$

$$R = \frac{v_0^2}{g} \sin 2\theta_0$$

$$T = \frac{2v_0 \sin \theta}{g}$$

$$f = \mu N$$

$$D = \frac{1}{2} C \rho A v^2$$

$$W = \vec{F} \cdot \vec{d} = Fd \cos \phi$$

$$F = -kx$$

$$W = -\frac{1}{2}kx^2$$

$$P_{avg} = \frac{W}{\Delta t}$$

$$P = \frac{dW}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$U = \frac{1}{2}kx^2$$

$$x_{com} = \frac{1}{M} \sum_{i=1}^n m_i x_i$$

$$\vec{r}_{com} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

$$\vec{p} = m\vec{v}$$

$$\vec{F}_{net} = \frac{d\vec{p}}{dt}$$

$$\Delta \vec{p} = \vec{J}$$

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}(t) dt$$

$$J = F_{avg} \Delta t$$

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i}$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$$

$$\theta = \frac{s}{r}$$

$$\omega_{avg} = \frac{\Delta \theta}{\Delta t}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha_{avg} = \frac{\Delta \omega}{\Delta t}$$

$$\alpha = \frac{d\omega}{dt}$$

$$\omega = \omega_0 + \alpha t$$

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega + \omega_0)t$$

$$\theta - \theta_0 = \omega t - \frac{1}{2} \alpha t^2$$

$$s = \theta r$$

$$v = \omega r$$

$$a_t = \alpha r$$

$$a_r = \frac{v^2}{r} = \omega^2 r$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega}$$

$$I = \sum m_i r_i^2$$

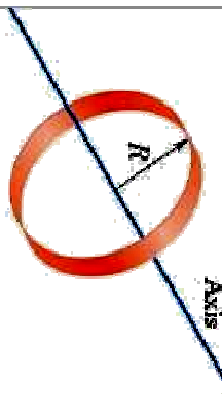
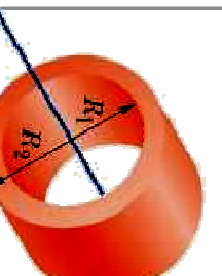
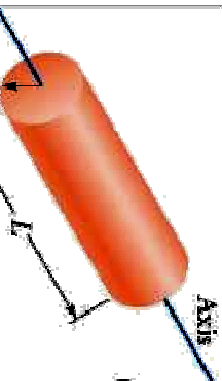
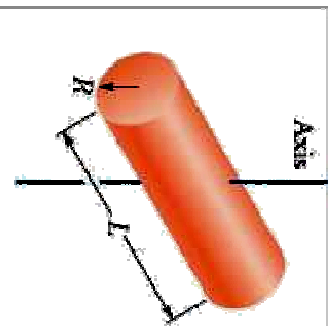
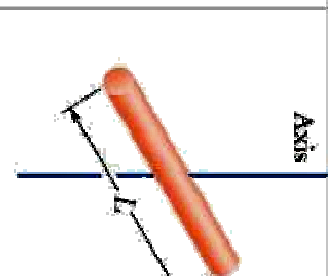
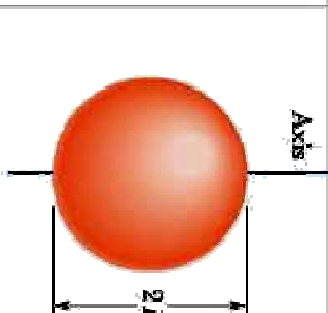
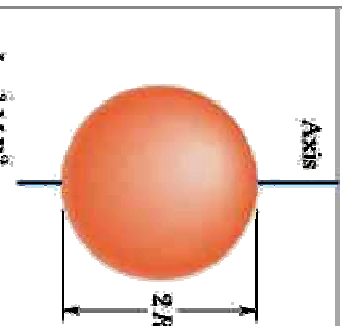
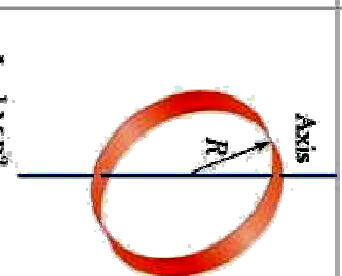
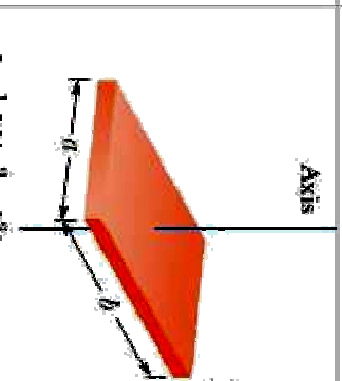
$$I = \int r^2 dm$$

$$I = I_{com} + Mh^2$$

$$\tau = rF \sin \phi$$

$$\tau = I\alpha$$

TABLE 10-2

 Hoop about central axis $I = MR^2$ (a)	 Annular cylinder (or ring) about central axis $I = \frac{1}{2}M(R_1^2 + R_2^2)$ (b)	 Solid cylinder (or disk) about central axis $I = \frac{1}{2}MR^2$ (c)
 Solid cylinder (or disk) about central diameter $I = \frac{1}{2}MR^2 + \frac{1}{12}ML^2$ (d)	 Thin rod about axis through center perpendicular to length $I = \frac{1}{12}ML^2$ (e)	 Solid sphere about any diameter $I = \frac{2}{5}MR^2$ (f)
 Thin spherical shell about any diameter $I = \frac{2}{3}MR^2$ (g)	 Hoop about any diameter $I = \frac{1}{2}MR^2$ (h)	 Slab about axis through center perpendicular to face $I = \frac{1}{12}M(a^2 + b^2)$ (i)